

Linear systems and elimination

Sharp growth for unequal lower and upper bandwidths

Difficulty: challenging **Importance:** interesting to specialist**Status:** [Lean verified](#)

Independently reviewed resolution - 2026-09-11

Sharp growth classification. Theorem 1 and Sections 2-4 prove $G(0, q) = 1$ and $G(p, q) = h_{p+q}$ for $p \geq 1$, where $h_t = 0$ for $t \leq 0$ and $h_t = 1 + \sum_{r=1}^p h_{t-r}$ otherwise. The upper bound covers every complex input and admissible tie path, with growth over all active entries in the fixed original ordering. A real nonsingular matrix of order $2p + q + 1$ attains it, including zero upper bandwidth.

Author: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. [Complete manuscript](#), [independent proof review](#), and [submission record](#). The supplied notes were reconstructed with substantial AI assistance. That September 2026 informal audit was independent agent verification; it did not supply external human peer review or formal proof-assistant certification. The later Lean verification is recorded separately below. No novelty or priority claim is made.

The difficulty, importance and rating rationale below are historical assessments of the original open target. Original statements, references and dated audits are preserved.

Rating rationale: Challenging reflects interacting fill-in and pivot choices across arbitrary unequal bandwidths; specialist impact is a sharp stability classification for banded elimination.

Lean proof and verification evidence

Formalization and verification submission: George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology. **Original mathematical proof:** Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. The [complete Lean proof](#) formalizes the full retained target.

The proof covers every original unequal natural bandwidth pair, every allowed positive dimension, all nonsingular complex banded inputs in their given ordering, and every legal maximal-modulus pivot tie. Its bound includes every active entry and stage. A nonsingular rational family attains the bound, including zero upper bandwidth; the identity case handles zero lower bandwidth. The final [sharp theorem](#) proves both the greatest attained value and the actual real supremum. The [28 declarations](#), [independent Challenge](#), [definitions](#) and [source correspondence](#) record their exact meaning.

On **16 September 2026 UTC**, [non-root Linux run 35081003513](#) checked literal proof commit `032d4c86c52ffde0c4d440f28527ba43555a0a24`. All 28 Comparator statement matches, Lean default-kernel replay, LeanCert kernel-trust assertions and required rejection/sandbox controls passed. Transitive axioms are restricted to *propext*, *Classical.choice* and *Quot.sound*. The [actual evidence](#) authenticates all 95 inputs and the complete 27-file proof. Lean 4.33.1, Mathlib and LeanCert are pinned. Only an exact half-bound uses a numerical certificate; dimensions, recurrences and powers remain symbolic, with no interval subdivision.

Two independent AI-agent [source/runtime reviews](#) support this record. Substantial OpenAI Codex assistance is disclosed. External human peer review and checker/platform infallibility are not claimed. The [project README](#) supplies reproduction commands. The named proof run is distinct from later publication and upstream merge runs, which require separate checks. A [bounded public audit](#) found no competing target-named formalization in its recorded snapshot; this adds verification of the existing resolution and makes no new mathematics or priority claim.

Problem statement

Work in exact arithmetic over $\mathbb{C}$. For integers $p, q \geq 0$, define

$$\mathcal{B}_n(p, q) = \{A \in \mathbb{C}^{n \times n} : \det A \neq 0, \quad a_{ij} = 0 \text{ if } i - j > p \text{ or } j - i > q\}.$$

Gaussian elimination with partial pivoting (GEPP) chooses a largest-modulus entry in the active first column, moves its row to the first position, and forms the trailing Schur complement. For a permitted path π , let $S_1 = A, \dots, S_n$ be its active matrices and define

$$\rho(A, \pi) = \frac{\max_{1 \leq k \leq n} \|S_k\|_{\max}}{\|A\|_{\max}}, \quad \|M\|_{\max} = \max_{i,j} |m_{ij}|.$$

Determine, for all $p \neq q$, the sharp bound independent of dimension,

$$G(p, q) = \sup_{n \geq 1 + \max(p, q)} \sup_{A \in \mathcal{B}_n(p, q)} \sup_{\pi \text{ permitted by GEPP}} \rho(A, \pi).$$

Identify the least universal upper bound and matching examples approaching it. All tie choices are included. The bandwidth restrictions apply in the given ordering; no reordering before GEPP is allowed. Bandwidths are *at most* p and q ; unequal pairs with one zero bandwidth are included. For $p \geq 1$, the equal-bandwidth case is known:

$$G(p, p) = 2^{2p-1} - (p-1)2^{p-2}.$$

This asks how unequal bandwidths affect worst-case element growth in banded linear solves. All unequal pairs constitute one problem.

References

N. J. Higham, *Accuracy and Stability of Numerical Algorithms*, second edition, SIAM (2002), Problem 9.15(a), p. 193; definitions and Theorem 9.11, pp. 172–173. Z. Bohte, *Bounds for Rounding Errors in the Gaussian Elimination for Band Systems*, J. Inst. Maths. Applics. 16 (1975), 133–142.

Earlier status check — 2026-09-08

The book’s problem was checked directly. Its wording omits the field; this entry adopts $\mathbb{C}$ from its referenced Theorem 9.11. Searches for *Higham 9.15 growth, growth factor lower bandwidth upper bandwidth*, and *Bohte Gaussian 1975* found no general unequal-bandwidth solution. Shah–Urschel’s [August 31, 2026 revision](#), Theorems 2.2–2.3, studies sparsity constraints without determining this extremal function. No recent explicit reaffirmation of this particular question’s openness was located; the status rests on the original question and this bounded later-literature check.

Audit update — 2026-09-10

Higham’s [Problem 9.15\(a\)](#), p. 193, remains the explicit historical source. Unequal-bandwidth and sharp-growth searches, including the [August 2026 complete-pivoting manuscript](#), did not identify a sharp answer for this partial-pivoting sparsity pattern. This is a bounded historical-source assessment, without a recent explicit reaffirmation.